

Brain-body integromics of the ayahuasca experience

Francisco Madrid-Gambin^{a,*}, Pablo Mallaroni^b, Noemí Haro^a, Oscar J. Pozo^a,
Natasha L. Mason^b, Johannes T. Reckweg^b, Lilian Kloft-Heller^b, Kim van Oorsouw^c,
Stefan W. Toennes^d, Johannes G. Ramaekers^{b,*}

^a Applied Metabolomics Research Group, Hospital del Mar Research Institute, Barcelona 08003, Spain

^b Department of Neuropsychology and Psychopharmacology, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, MD 6200, the Netherlands

^c Department of Clinical Psychological Science, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, MD 6200, the Netherlands

^d Goethe University Frankfurt, University Hospital, Institute of Legal Medicine, Frankfurt Main D-60596, Germany

ARTICLE INFO

Keywords:

Ayahuasca
Dimethyltryptamine
Connectome
Metabolomics
Psychedelics
Integromics

ABSTRACT

Ayahuasca is a psychoactive brew containing N,N-dimethyltryptamine (DMT) and β -carboline alkaloids that induces marked alterations in perception, emotion and self-referential processing. However, the multiscale biological organization linking peripheral metabolism, brain network dynamics, neurochemistry, and subjective experience in humans remains poorly understood. Here, we applied an integrative, within-subject, multiblock partial least squares framework to model coordinated changes across four complementary biological and phenotypic layers: plasma psychoactive alkaloids, targeted metabolomics, resting-state fMRI-derived functional connectomes, and multidimensional subjective experience assessed with the 5-Dimensional Altered States of Consciousness (5D-ASC) scale, in 20 experienced ceremonial ayahuasca users. Complementary ¹H-MRS data were used to examine associations between peripheral metabolism, posterior cingulate cortex neurochemistry, and default mode network (DMN)-related connectivity. Multilayer integration revealed that the experiential dimensions oceanic boundlessness, visionary restructuralization and auditory alterations covaried with circulating DMT and β -carbolines, alterations in lipid, amino acid and energy metabolisms and reconfiguration of dorsal attention- and DMN-related connectivity. Shared network features across experiential dimensions were most strongly associated with endocannabinoid-related N-acylethanolamines, acylglycerols, and ceramides, extending canonical serotonergic models toward downstream lipid-signalling and metabolic processes. Complementary rCCA analyses further showed structured covariation between peripheral metabolites, posterior cingulate cortex neurochemistry, and DMN-related connectivity. Together, these findings indicate that psychedelic states reflect coordinated, system-level interactions between peripheral metabolism and functional brain networks rather than isolated neurochemical or neural events. Framed within a brain-body integromics perspective, this work provides translationally relevant insight into metabolic pathways that may modulate brain function and subjective response, with potential implications for neuropsychiatric and pharmacometabolic research.

1. Introduction

Ayahuasca is a complex psychoactive brew traditionally prepared

from *Banisteriopsis caapi* and *Psychotria viridis*, containing β -carboline alkaloids (harmine, harmaline, tetrahydroharmine) and the serotonergic psychedelic N,N-dimethyltryptamine (DMT) that produces profound

Abbreviations: 2-AG, 2-arachidonoyl glycerol; 2-OG, 2-oleoylglycerol; ASC, 5-Dimension Altered States of Consciousness Rating Scale; 5HIAA, 5-hydroxyindoleacetic acid; AAA, aromatic amino acid; AEA, anandamide; BCAA, branched-chain amino acid; AUD, auditory alterations; Cer, ceramide; DGLA, dihomogamma-linolenylethanolamide; DMN, default mode network; DA, dorsal attention; DAG, false discovery rate, diacylglycerol; DEA, docosatetraenylethanolamide; DHEA, docosahexaenylethanolamide; FDR, frontoparietal; FP, hexosylceramides; fMRI, functional magnetic resonance imaging; HexCer, hexosylceramide; MAG, monoacylglycerol; LNAA, large-neutral amino acid; LEA, linoleylethanolamide; L, limbic; LC-MS/MS, liquid chromatography–tandem mass spectrometry; LLOQ, lower limits of quantitation; LPC, lysophosphatidylcholine; PLS-DA, partial least squares discriminant analysis; DMT, N,N-dimethyltryptamine; SM, somatomotor; VAS, visual analogue scale; VA, ventral attention; VIS, visual; VUS, visionary restructuralization from the original German “*Visionäre Umstrukturierung*”.

* Corresponding authors.

E-mail addresses: fmadrid@researchmar.net (F. Madrid-Gambin), j.ramaekers@maastrichtuniversity.nl (J.G. Ramaekers).

<https://doi.org/10.1016/j.bioph.2026.119391>

Received 14 February 2026; Received in revised form 1 April 2026; Accepted 15 April 2026

Available online 23 April 2026

0753-3322/© 2026 The Author(s). Published by Elsevier Masson SAS. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

alterations of consciousness [1].

Pharmacokinetic studies show that DMT and β -carbolines rapidly reach measurable plasma levels after ingestion, with peak concentrations typically observed within the first 2 h [2]. In addition, DMT exposure and subjective effects vary as a function of differences in harmine alkaloid content [3]. These alkaloids interact not only with serotonergic receptors but also with broader metabolic networks, influencing (serotonergic) neurotransmitter precursors, amino acid balance, and endocannabinoid pathways [4]. Previous studies have shown that the psychedelic effects of ayahuasca are largely mediated by 5-HT_{2A} receptor activation [5,6], while downstream biochemical cascades involving monoaminergic and lipid signalling are less well understood. In addition to MAO-A inhibition, β -carbolines may exert additional monoaminergic effects; for example, tetrahydroharmine has been reported to inhibit the serotonin transporter (SERT), potentially enhancing extracellular serotonin levels [7,8]. Consistent with this broader pharmacological profile, blockade of 5-HT_{2A} receptors with ketanserin attenuated, but only partially blocked, the subjective effects of ayahuasca, suggesting that its psychoactive effects are not mediated exclusively by 5-HT_{2A} signalling [9,10]. Thus, the overall pharmacological action of ayahuasca likely reflects the combined and interacting effects of DMT-mediated receptor activation and β -carboline-mediated modulation of monoaminergic neurotransmission. Beyond single profiling, strategic use of metabolomic dynamics enables the interrogation of coordinated biochemical processes and their coupling with neural and experiential states [11,12].

Parallel advances in resting-state functional magnetic resonance imaging (fMRI) neuroimaging have shown that ayahuasca, similar to other serotonergic psychedelics, alters large-scale functional connectivity (FC). Acute psychedelic states are characterized by a desynchronization of FC and increased neural signal complexity (or neural entropy), accompanied by a shift from segregated network organization—characterized by reduced modularity and diminished within-network connectivity—to more globally integrated patterns of communication [13–16].

Therefore, psychedelic effects span molecular, neural, and experiential domains, rendering single-level analyses insufficient to capture their systems-level biological impact. A multiscale integrative framework is therefore necessary to explain how peripheral biochemical perturbations translate into large-scale brain network reorganization and subjective experience. This approach is particularly well suited to ayahuasca, a complex psychoactive mixture containing multiple interacting compounds, for which integrating plasma alkaloid levels and metabolomics with functional connectomics enables direct links between systemic biochemical dynamics and brain network activity during the psychedelic state. The present study integrates multiple layers of data, i.e. metabolomics, connectomics, plasma concentrations of DMT and harmine alkaloids, and subjective ratings of the psychedelic experience, that were collected from Santo Daime members assessed at baseline and after ayahuasca ingestion. The main purpose of the study was to map how specific metabolic changes correlate with alterations in brain connectivity and how these, in turn, correspond to differential aspects of the psychedelic experience during ayahuasca. Previous publications on ayahuasca ingestion presented subjective experience, amino acid metabolites, alkaloid concentrations and functional connectivity measures in isolation [4,17,18]. By jointly integrating biochemical, metabolic, neurofunctional, and experiential data, we aim to elucidate the multiscale dynamics underlying the psychedelic state. This integrative framework not only contributes to the neurobiology of ayahuasca but also supports the development of systems-level models of altered states of consciousness.

2. Materials and methods

2.1. Subjects and study design

A total of 24 healthy adult volunteers participated in a within-subject, fixed-order, observational study. Data of 20 participants (11 males, 9 females; mean age = 37.1 ± 10.6 years) with complete imaging and metabolomics datasets entered data analysis. Participants were long-term members of the Dutch Santo Daime community, ensuring prior experience with ayahuasca. Assessments were conducted on two consecutive days. On Day 1, assessments were obtained at baseline (non-intoxicated state). On Day 2, assessments were obtained following ayahuasca ingestion (mean = 24 mL, *SD* = 8.16). Inter-individual variability in systemic exposure was addressed through pharmacokinetic assessment, by quantifying plasma concentrations of DMT and β -carbolines. Sessions were conducted individually and spaced approximately 90 min apart to minimize overlap of acute effects across participants. Testing was organized in four laboratory cycles of six participants each, with each participant assessed at the same time on both days to control for circadian variation. Exclusion criteria included the presence of ferromagnetic implants, pregnancy, and the use of medications within 24 h prior to testing. A full inclusion criterion and testing procedures are detailed in the [Supplementary Materials](#).

This study was conducted in accordance with the Declaration of Helsinki and its 2013 revision in Fortaleza, Brazil, and adhered to the Medical Research Involving Human Subjects Act (WMO). Ethical approval was obtained from the Medical Ethics Committee of the Academic Hospital and University of Maastricht (NL70901.068.19 / METC19.050). All participants provided informed consent after being fully briefed on study procedures, potential risks, and their right to withdraw without penalty.

2.2. Psychedelic state questionnaire

Subjective effects were assessed using the 5-Dimensional Altered States of Consciousness Questionnaire (5D-ASC) [19–21]. This 94-item self-report instrument captures five core domains of altered states: oceanic boundlessness (OB), visionary restructuring (VUS, from the original German “*Visionäre Umstrukturierung*”), anxious ego dissolution (AED), auditory alterations (AUD), and reduction in vigilance (VIR). Participants completed the questionnaire at the end of the second day by rating each item on a 10-cm visual analogue scale (VAS) anchored from 0% (“No, not more than usual”) to 100% (“Yes, much more than usual”).

2.3. Functional MRI imaging

The present study reuses data previously reported [17], but applies a novel data processing and integrative framework. Participants completed one imaging session per visit to collect resting-state fMRI data, approximately 60 min after dosing. To minimise circadian effects, all imaging sessions were conducted at the same time of day for each participant across visits. Images were acquired on a Siemens MAGNETOM 7 T MRI scanner. For complete details on image acquisition procedures and preprocessing, see [Supplementary Materials](#).

BOLD timeseries were preprocessed using an in-house pipeline based on established procedures [17,22–24], including motion correction, voxel-wise nuisance regression, and temporal filtering, incorporating aCompCor-based removal of physiological noise [25]. Head motion was assessed using framewise displacement (FD), the temporal derivative of BOLD signal variance (DVARS) [26], and signal standard deviation metrics, with participants exhibiting mean FD > 0.5 mm excluded from further analyses. Volumes affected by excessive head motion were identified and censored (“scrubbed”) when framewise displacement (FD) exceeded 0.5 mm or when DVARS standardized by its temporal standard deviation exceeded the threshold defined as the 75th percentile

plus 1.5 interquartile ranges. Three participants were excluded due to excessive motion. In the remaining sample, no significant differences in head motion were observed (see [Supplementary Figure 1](#)).

Denoised BOLD time series were parcellated into 200 cortical ROIs using the 2 mm Schaefer atlas [27] and mapped to native EPI space. Static functional connectivity was computed as pairwise Pearson correlations between regions and Fisher z-transformed. To reduce dimensionality, connectivity values were averaged within Yeo et al.'s seven canonical resting-state networks [28], including visual (VIS), somatomotor (SM), dorsal attention (DA), ventral attention (VA), limbic (L), frontoparietal (FP), and default mode network (DMN), retaining only unique network-level features for subsequent analyses.

2.4. Magnetic resonance spectroscopy (MRS) acquisition

Proton magnetic resonance spectroscopy (^1H -MRS) data were acquired in all participants to quantify in vivo brain metabolite concentrations between 70- and 90-minutes post-dosing. Single-voxel spectra were obtained from the posterior cingulate cortex and visual cortex using a stimulated echo acquisition mode (STEAM) sequence (TE = 6.0 ms, TR = 5.0 s, 64 averages) [29] on a Siemens MAGNETOM 7 T MRI scanner. Spectroscopic voxels were positioned by a trained operator. MRS outcome measures consisted of concentration ratios of aspartate, N-acetylaspartate (NAA), glutamate, N-acetylaspartylglutamate (NAAG), GABA, glutamine, taurine, glycerophosphocholine, phosphocholine, glutathione, glycine, and myo-inositol relative to total creatine (creatine + phosphocreatine). The use of total creatine as an internal reference minimizes variability arising from transmit/receive RF inhomogeneity, magnetic field drift, and cerebrospinal fluid (CSF) partial volume effects within the voxel [30].

2.5. Plasma alkaloid concentrations

The present study is based on data previously reported [18]. A blood sample was taken at 60 min post drinking to determine alkaloid concentrations in plasma. Plasma aliquots were frozen at -80°C until analysis. Alkaloid concentrations were determined in 200 μL plasma using liquid chromatography–tandem mass spectrometry (LC–MS/MS) (Agilent, Waldbronn, Germany) after ethyl acetate extraction. Calibration curves covered the range 0.25 – 40 ng/mL with lower limits of quantitation (LLOQ) of DMT 0.077, harmine 0.13, harmaline 0.23, and tetrahydroharmine 0.18 ng/mL.

2.6. Metabolomics analysis

Plasma samples were taken 30 min after the MRI scanning session (i.e. 150 min after ayahuasca ingestion) on both the baseline and ayahuasca study days. Following protein precipitation, samples were analyzed using six targeted LC–MS/MS methods covering distinct metabolite classes, as previously described [31–36]. A set of 157 targeted biomarkers, composed of 12 compounds related to endocannabinoids, 25 markers related to amino acid metabolism, 13 steroids, 25 markers related to energy metabolism, 26 acylglycerols, 25 ceramides, 18 lysophosphatidylcholines (LPCs) and, 6 compounds of choline metabolism; were determined by selected reaction monitoring by LC–MS/MS system consisting of an Acquity UPLC instrument (Waters Associates, Milford, MA, USA) coupled to a triple quadrupole (TQS Micro, Waters) mass spectrometer. MassLynx software V4.1 (Waters Associates) was used for peak integration and data management. Large neutral amino acids (LNAA) were calculated as the sum of valine, leucine, isoleucine, tyrosine, tryptophan, and phenylalanine. Extended details are published elsewhere [4].

2.7. Statistical analysis

Demographic data were summarized as mean $\pm$ SD, counts, or

percentages. All analyses were conducted in R version 4.4. Features with more than 80% missing values were excluded. Data distributions were evaluated both across data blocks and within individual features, and normality was assessed using Shapiro–Wilk tests. To assess the effects of ayahuasca on metabolite levels, fold changes were calculated and paired comparisons between baseline and post-ingestion values were performed using Wilcoxon signed-rank tests. For downstream integrative analyses, change scores (Δ) were computed for alkaloids, metabolomic variables, functional connectivity measures, and subjective experience by subtracting baseline values from post-administration values within each dataset.

Additionally, generalized linear models with a Poisson distribution and log link function were applied to examine associations between metabolite changes (Δ) and subjective experience. Predictor variables consisted of Δ values (post-ingestion minus baseline), and models were fitted separately for each of the surviving 5D-ASC dimensions. Multiple comparisons were controlled using the Benjamini–Hochberg false discovery rate (FDR) procedure.

The main data integration was conducted using the DIABLO algorithm (Data Integration Analysis for Biomarker discovery using Latent cOmponents) [37], implemented in the mixOmics R package [38]. A multiblock partial least squares (PLS) model was implemented, with the 5D-ASC dataset defined as the response block (Y-block). For model training and validation, autoscaled datasets were partitioned into training (75%) and testing (25%) subsets, with the 5D-ASC dataset specified as the response block (Y-block). The inter-block correlation in the design matrix was set to 0.5 to balance between discrimination and integration [39]. Singular Value Decomposition (SVD) was independently applied to each block. The optimal number of latent components of each layer was selected by minimizing the root-mean-square error (RMSE) of predictions on the test set, using a modified implementation of the “predict.block.pls” function.

Feature selection was conducted using a bootstrapping procedure, in which datasets were randomly shuffled and the multiblock-PLS model recalculated at each iteration. Statistical significance was determined by comparing the resulting similarity scores to a null distribution generated from 5000 permutations. The Benjamini–Hochberg procedure was used to control the false discovery rate (FDR), and features with an FDR-adjusted p-value < 0.05 were considered statistically significant. A network graph was generated to visualize significant features and their interrelationships across data layers. Additionally, an ASC-centred core network was generated, including only nodes connected to at least one ASC variable and simultaneously linked to nodes in each of the remaining layers (quadrupartite multi-layer network). [Fig. 1](#) illustrates the overall study design and data integration framework.

In addition to the main modelling, regularized canonical correlation analysis (rCCA) was applied to explore associations involving an additional data layer, namely brain neurochemistry measured in two regions, in relation to peripheral metabolism and functional connectivity, using the mixOmics R package. Three separate rCCA models were implemented using delta-data (baseline-subtracted values), including (i) plasma metabolomics and posterior cingulate cortex MRS metabolites, (ii) posterior cingulate cortex MRS metabolites and DMN-related functional connectivity measures, and (iii) visual cortex MRS metabolites and visual network connectivity measures. Prior to analysis, all variables were mean-centered and scaled to unit variance. Regularization parameters were optimized using a grid search procedure with leave-one-out cross-validation, testing shrinkage values ranging from 0.001 to 1. Optimal parameters were selected based on cross-validated performance and subsequently used to fit the final rCCA models. For each analysis, pairwise correlation matrices were derived to quantify the strength of association between variables from the paired data blocks. These matrices reflect patterns of cross-block covariation captured by the canonical components and were interpreted as measures of statistical association rather than causal relationships.

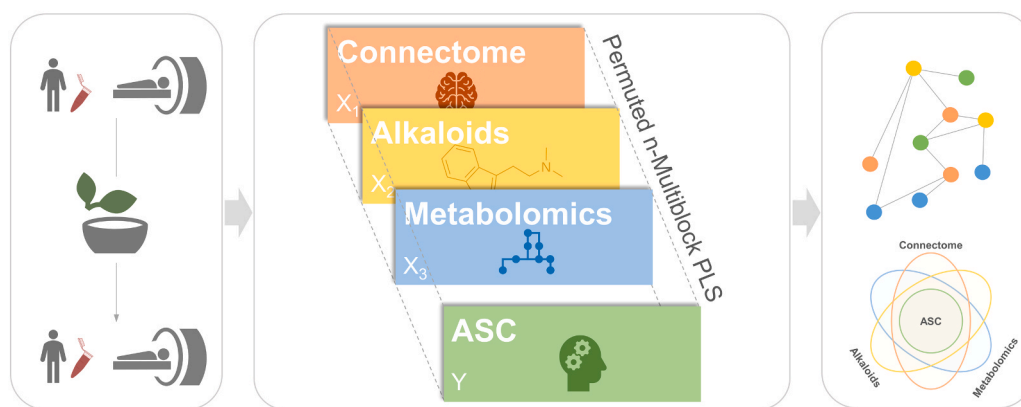


Fig. 1. Study design and multi-block data integration framework using latent components. The left panel shows a schematic of the study design, in which blood samples and fMRI data were collected from participants before and after ayahuasca consumption. The middle panel outlines the integrative analytical pipeline incorporating four data layers: brain connectivity (connectome), plasma alkaloids, global metabolomic signatures, and psychometric assessments of altered states of consciousness (ASC). Multi-block integration was performed using a latent component model within a cross-validated framework to capture shared variance across modalities. The right panel presents the resulting multiscale networks linking neurobiological, biochemical, and experiential dimensions across all blocks (upper) and at the centre of overlap (lower).

3. Results

Of the 24 participants enrolled, 20 completed the full study protocol, including blood sampling, resting-state fMRI, plasma metabolomic analysis, and subjective ratings of the psychedelic experience assessed with the 5D-ASC questionnaire. Three participants were excluded because of excessive head motion during fMRI, and one was excluded due to insufficient biological samples. Demographic and lifestyle characteristics of the final sample are summarized in [Supplementary Table 1](#). Participants self-administered a single volume of ayahuasca equivalent to their usual dose (mean: 24 mL, SD: 8.16), prepared from a single batch by the Church of Santo Daime. The brew contained 0.14 mg/mL DMT, 4.50 mg/mL harmine, 0.51 mg/mL harmaline, and 2.10 mg/mL tetrahydroharmine. At the metabolic level, ayahuasca ingestion was associated with a marked modulation of the endocannabinoid-related N-acyl ethanolamine panel, with significant increases in anandamide (AEA), oleoylethanolamide (OEA), palmitoleylethanolamide (POEA), and dihomog- γ -linolenylethanolamide (DGLA) after FDR correction ([Supplemental Table 2](#)). More limited pre-post effects were also detected in other metabolic blocks, including amino acid-related indices such as 5HIAA/5HT and leucine/LNAA ([Supplemental Table 3](#)), as well as cortisone-related steroids, particularly 20 α -dihydrocortisone and cortisone ([Supplemental Table 4](#)). By contrast, no plasma acylglycerol or ceramide/hexosylceramide measure survived FDR correction in the pre-post comparison, indicating that ayahuasca did not induce robust group-level changes in these lipid classes ([Supplemental Table 5 and 6](#)).

3.1. Metabolic univariate associations with ASC

We next examined whether interindividual metabolic responses to ayahuasca, using delta-data, were associated with the three ASC dimensions retained in the integrative model. In the endocannabinoid-related N-acyl ethanolamine block, DGLA and LEA were positively associated with OB, AUD, and VUS, while POEA, OEA, and AEA showed additional positive associations with selected ASC dimensions ([Supplementary Table 8](#)). In the amino acid block, AUD and VUS were inversely associated with several branched-chain and large neutral amino acid measures, particularly leucine, isoleucine, tryptophan, phenylalanine, LNAA, and phenylalanine/LNAA, whereas glutamate and creatine were positively associated with OB, and creatine was also positively associated with VUS ([Supplementary Table 9](#)). In the energy metabolism block, several metabolites and metabolite ratios were significantly associated with at least two of the three main ASC dimensions, including fumarate, fumarate/succinate, isocitrate,

isocitrate/citrate, citrate/malate, lactate/pyruvate, and glutarate ([Supplementary Table 10](#)). Notably, despite the absence of robust group-level changes in acylglycerols and ceramide-related lipids after ayahuasca ingestion, several lipid Δ -values were significantly associated with ASC scores. Positive associations of MAG/DAG 18:2 and MAG/DAG 16:0, and negative associations of MAG/DAG 18:0, were observed across the ASC dimensions ([Supplementary Table 12](#)). Ceramide- and hexosylceramide-related changes showed predominantly inverse associations, especially with VUS ([Supplementary Table 13](#)).

3.2. Multiblock model optimization

For training the model, the optimal number of components in the multiblock-PLS analysis was determined by RMSE of the 5D-ASC dataset predictions in the test set. This optimization process identified one latent component as optimal for modelling with the lowest RMSE in the validation set ([Supplementary Figure 2](#)). In the validation set, this component showed that AUD, VUS and OB from the 5D-ASC block had the highest loading values before the inflection point ('elbow'), along with tetrahydroharmine, DMT and harmaline from the alkaloids block, and several monoacylglycerol (MAG) and diacylglycerol (DAG) species, such as MAG 19:0, MAG 18:2 (2-linoleoyl glycerol), DAG 18:2, DAG 18:0 18:2 and MAG 20:4, also known as the endocannabinoid 2-arachidonoylglycerol (2-AG), in the metabolomics block, as shown in [Supplementary Figure 3](#). In this component, other endocannabinoid-related N-acyl ethanolamines appeared such as dihomog- γ -linolenylethanolamide (DGLA), linoleylethanolamide (LEA) and oleoylethanolamide (OEA); serotonin (5-HT) and ceramides-related markers such as HexCer/Cer 20:0. Regarding the connectome edges, DA-DA, SM-DA, DA-DMN and SM-L showed the highest loading values, but a sharp decline of loadings values was not observed in this block. The retained first latent component of the final model explained 47.1% of the variance in the 5D-ASC block, providing additional context for the contribution of the multivariate model to individual differences in the acute subjective effects of ayahuasca. The same component accounted for 82.9% of the variance in alkaloids, 28.9% in functional connectivity, and 9.0% in metabolomics, supporting the interpretation that the principal multiblock axis was driven mainly by shared variation between exposure-related alkaloids, connectomic changes, and altered states of consciousness, with a smaller contribution from plasma metabolomic variation.

3.3. Multi-layered network of cross-block associations

Permutation test-based feature selection denoted that from 6340

combinations, only 610 remained significant after adjustment for multiple comparisons. Our multiblock integration model revealed that the four data layers (plasma alkaloids, metabolomics, functional connectome, and the psychedelic experience) were strongly interrelated (Fig. 2). The Circos plot (Fig. 2A) illustrates the overall correlation structure across blocks, showing that only three 5D-ASC dimensions—OB, VUS, and AUD—were significantly represented in the integrated network. Among the biochemical variables, all quantified alkaloids (DMT, harmine, harmaline, and tetrahydroharmine) exhibited the highest connectivity degree across the datasets, occupying a central hub position that bridged the peripheral biochemical and central nervous system domains (Fig. 2B). An interactive HTML version of the network, including all variable names and connection details, is available in the [Supplementary Material](#). Connectome-derived features followed in importance, with the most interconnected pairs being DA–DMN and VA–DMN, while SM–DA and DMN–DMN occupied topologically central positions in the integrated network, suggesting a coordinating role in the cross-domain associations. Their central importance was further indicated by the finding that edges associated with the DMN and DA networks were overrepresented relative to other connectome features. Within the metabolomics block, lipid-related pathways showed the strongest contribution, particularly acylglycerols and selected ceramides, which displayed extensive connectivity with alkaloid, functional connectivity and experiential features. Other metabolite families, including endocannabinoid-related N-acyl ethanolamines, amino acid derivatives, and energy metabolism intermediates, also exhibited relevant associations but were topologically distinct, clustering in a region of the network opposite to the ASC dimensions.

The ASC dimensions VUS, AUD, and OB exhibited the strongest

positive similarity scores with connectivity pairs involving DA and SM, particularly DA–DA and SM–DA, as well as the formation of long-chain hexosylceramides (HexCer/Cer 20:0 and HexCer/Cer 18:0) and all measured plasma alkaloids (all showing similarity scores > 0.5, FDR-corrected $p < 0.05$). Conversely, OB, VUS, and AUD—together with the alkaloids—were inversely associated with DA–DMN connectivity and a subset of acylglycerols, including MAG 12:0, MAG 14:0, MAG 18:2, MAG 19:0, MAG 20:4 (2-AG), and DAG 18:0/18:2 ($|\text{similarity score}| > 0.5$ for all cases, FDR-corrected $p < 0.05$).

Connectivity pairs such as L–DMN and DMN–DMN displayed similar but weaker negative associations ($|\text{similarity score}| > 0.4$, FDR-corrected $p < 0.05$), indicating a graded reorganization of intrinsic brain networks. N-acyl ethanolamines-based endocannabinoid anandamide (AEA) and endocannabinoid-related N-acyl ethanolamines, including, dihomono- γ -linolenylethanolamide (DGLA), oleylethanolamide (OEA), and linoleylethanolamide (LEA), positively correlated with all alkaloids, DA–DA, SM–DA, DMN–DMN, and L–DMN, while showing inverse correlations with DA–DMN, VA–DMN, and SM–L ($|\text{similarity score}| > 0.5$ for all cases, FDR-corrected $p < 0.05$). A comparable pattern was observed for amino acid metabolism: glutamine, glutamate, phenylalanine, and betaine showed robust correlations with alkaloid levels and functional connectivity features (e.g., DA–DA and DA–DMN). Peripheral 5-HT displayed additional associations with SM–DMN, SM–L, and L–DMN, segregating from the cluster of the aforementioned amino acids.

Among energy metabolism intermediates, citrate/malate, alpha-ketoglutarate, succinate/alpha-ketoglutarate, isovalerate, malonate, and mevalonate were positively correlated with DA–DA, SM–DA, and DMN–DMN connectivity, and inversely correlated with DA–DMN and

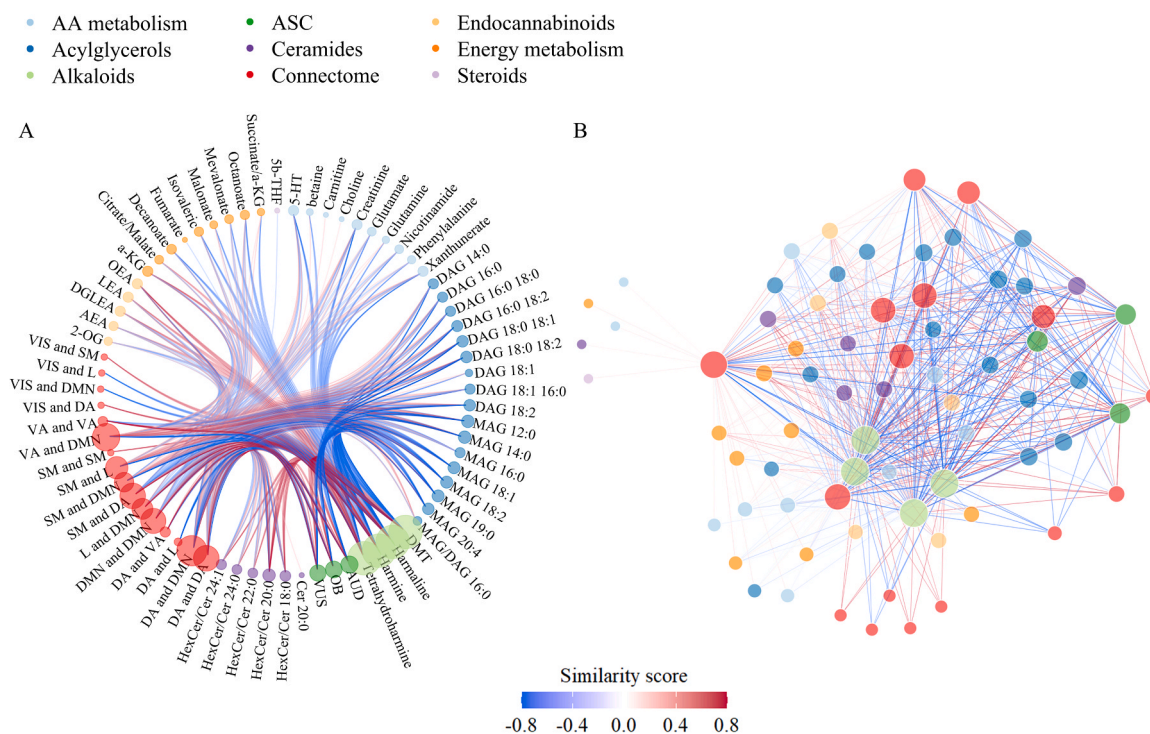


Fig. 2. Multi-layer network of cross-block associations derived from the integrative multiblock-PLS model. (A) Circos plot visualizing all significant associations (FDR-corrected $p < 0.05$, based on permutation testing), with variables grouped by functional family. Metabolomic features are organized into major pathways, including endocannabinoids, ceramides, acylglycerols, amino acid metabolism, energy metabolism, and steroids, as detailed in [Supplementary Table 1](#). Nodes represent individual variables, scaled by degree (number of cross-block connections), while edge colours denote the direction of association (red = positive, blue = negative), with line intensity reflecting the strength of correlation. (B) Integrative network plot displaying the overall topology and distribution of interconnected features. An interactive HTML version of the network, including all variable names and connection details, is available in the [Supplementary Material](#). Abbreviations: 2-OG, 2-oleoylglycerol; AEA, anandamide; AUD, auditory alterations; VUS, visionary restructuring; Cer, ceramide; DAG, diacylglycerol; DEA, docosahexaenylethanolamide; DGLA, dihomono- γ -linolenylethanolamide; DHEA, docosahexaenylethanolamide; DMT, N,N-dimethyltryptamine; DA, dorsal attention; EI, Elemental imagery; FP, frontoparietal; hexosylceramides; HexCer; MAG, monoacylglycerol; LEA, linoleylethanolamide; L, limbic; LNAA, large neutral amino acids; LPC, lysophosphatidylcholine; OB, oceanic boundlessness; OEA, oleylethanolamide; SM, somatomotor; VA, ventral attention; VIS, visual.

VA–DMN links, forming a cluster topologically opposite to the ASC cluster in the network representation.

3.4. ASC-centred core network

Next, we focused on the three ASC nodes as the central core connecting with the other three blocks, specifically, for each ASC node, we

retained only those nodes in the other three blocks that showed a correlation with that ASC node and also exhibited a connection to nodes in each of the remaining blocks.

Our findings highlighted a quadripartite link connecting the four data layers—alkaloids, metabolomics, functional connectivity, and ASC dimensions—through coherent cross-domain associations (Fig. 3A). For OB and VUS (Fig. 3B and D), the network in which features from the four

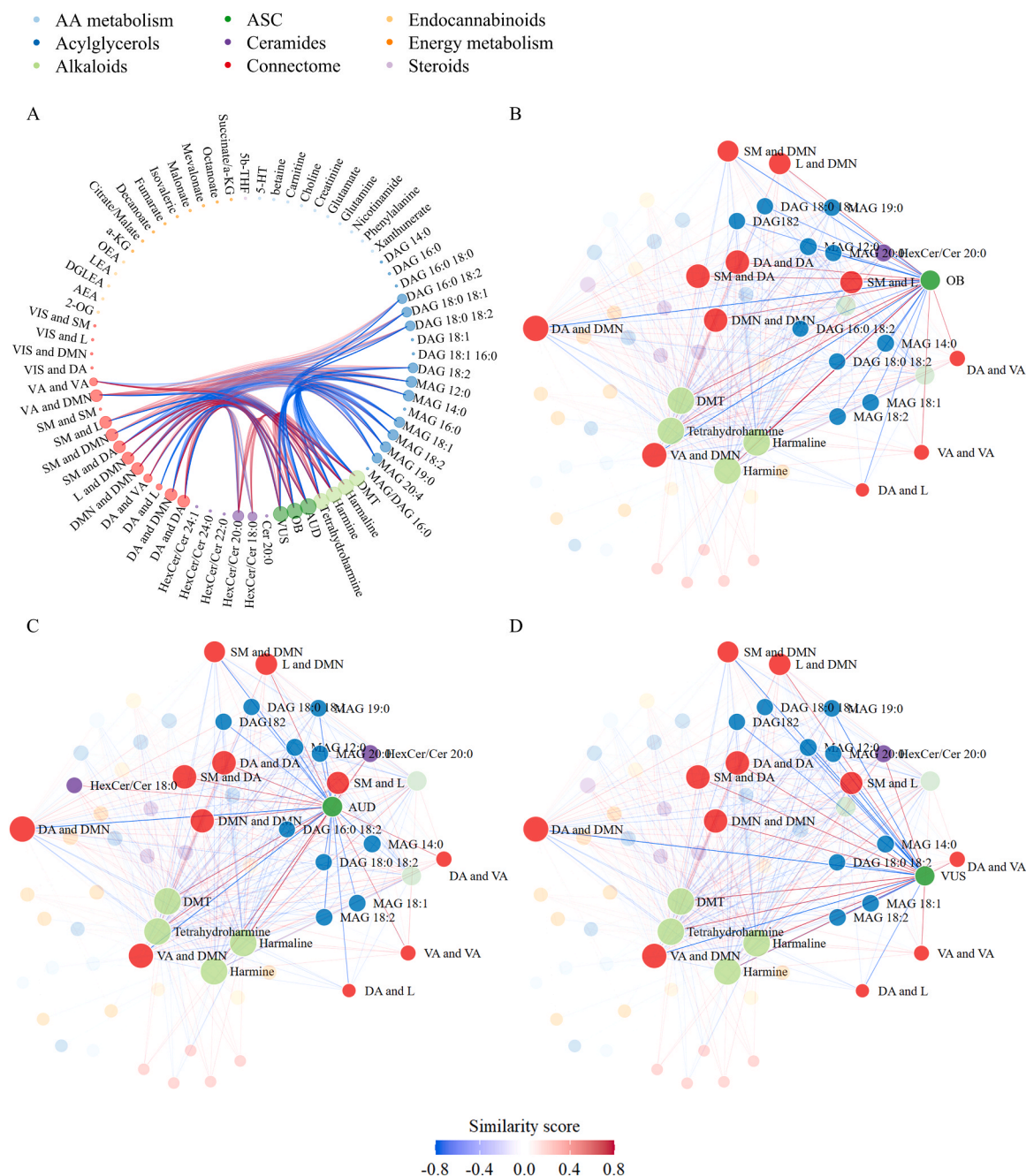


Fig. 3. ASC-centred core network. Quadripartite multi-layer network of cross-block associations derived from the integrative multiblock-PLS model. (A) Circos plot illustrating the quadripartite link across the four data layers: alkaloids, metabolomics, connectome, and ASC dimensions. Nodes represent individual variables, scaled by degree (number of cross-block connections), while edges indicate significant associations between layers. Panels B–D show integrative network plots highlighting the across-block associations with each of the ASC dimensions separately, revealing the topology of interconnected features. For clarity, metabolomic variables are grouped by compound family, as detailed in [Supplementary Table 1](#). Edge colours denote the direction of association (red = positive, blue = negative), with line intensity reflecting the strength of correlation. Node size represents the number of significant connections per variable. Abbreviations: 2-OG, 2-oleoylglycerol; AEA, anandamide; AUD, auditory alterations; VUS, visionary restructuring; Cer, ceramide; DAG, diacylglycerol; DEA, docosatetraenylethanolamide; DGLEA, dihomog- γ -linolenylethanolamide; DHEA, docosahexaenylethanolamide; DMT, N,N-dimethyltryptamine; DA, dorsal attention; EI, Elemental imagery; FP, frontoparietal; hexosylceramides; HexCer; MAG, monoacylglycerol; LEA, linoleylethanolamide; L, limbic; LNAA, large neutral amino acids; LPC, lysophosphatidylcholine; OB, oceanic boundlessness; OEA, oleylethanolamide; SM, somatomotor; VA, ventral attention; VIS, visual.

layers were intercorrelated simultaneously (overlapped features), included all plasma alkaloids, 10 acylglycerol and the formation of a HexCer lipid, and several connectome pairs mostly related either to DA, VA or DMN, including a prominent inverse association with DA-DMN. For AUD, the network included the same associations, and the formation of an additional HexCer (Fig. 3C). Interestingly, although endocannabinoids, amino acid metabolism, and energy metabolism displayed coordinated associations with plasma alkaloids and functional connectivity features, their relationships with the ASC dimensions (OB, VUS, and AUD) were only nominally significant, but did not survive corrections for multiple comparisons in the integrative model and were therefore excluded from the quadripartite network (see [Supplementary Table 14 and 15](#)). No significance was found for steroids, lysophosphatidylcholines and choline metabolism.

3.5. Associations with central neurochemistry

rCCA integrating plasma metabolomics with posterior cingulate cortex MRS metabolites revealed structured patterns of covariation between peripheral and central measures. Distinct clusters of plasma metabolites, including lipid species (e.g., mono- and diacylglycerols,

ceramides) and energy-related intermediates (lactate, malonate, mevalonate), showed differential associations with MRS-derived metabolites. MRS clusters 1 and 2, comprising creatine, aspartate, NAAG, and glutathione, were inversely associated with peripheral lipid species, whereas phosphocreatine (cluster 3) showed positive associations with these lipids. In addition, the plasma 5HIAA/serotonin ratio was positively associated with central myo-inositol, N-acetylaspartate, and choline-containing compounds (glycerophosphocholine + phosphocholine), shown in Fig. 4. In contrast, plasma neurotransmission-related compounds, including serotonin and amino acids, exhibited weaker and less specific associations with posterior cingulate cortex neurochemical measures. The corresponding regularized correlation matrix is provided in [Supplementary Table 16](#) (Fig. 5).

Integration of posterior cingulate cortex MRS metabolites with DMN-related functional connectivity measures revealed a structured organization of associations across connectivity patterns. Connectivity pairs involving within-network DMN and frontoparietal-DMN interactions exhibited similar association profiles across MRS variables, whereas those involving sensorimotor, visual, and dorsal attention networks showed an opposing pattern. MRS metabolites including NAA, NAAG, and glutamate/glutamine exhibited consistent associations with DMN-

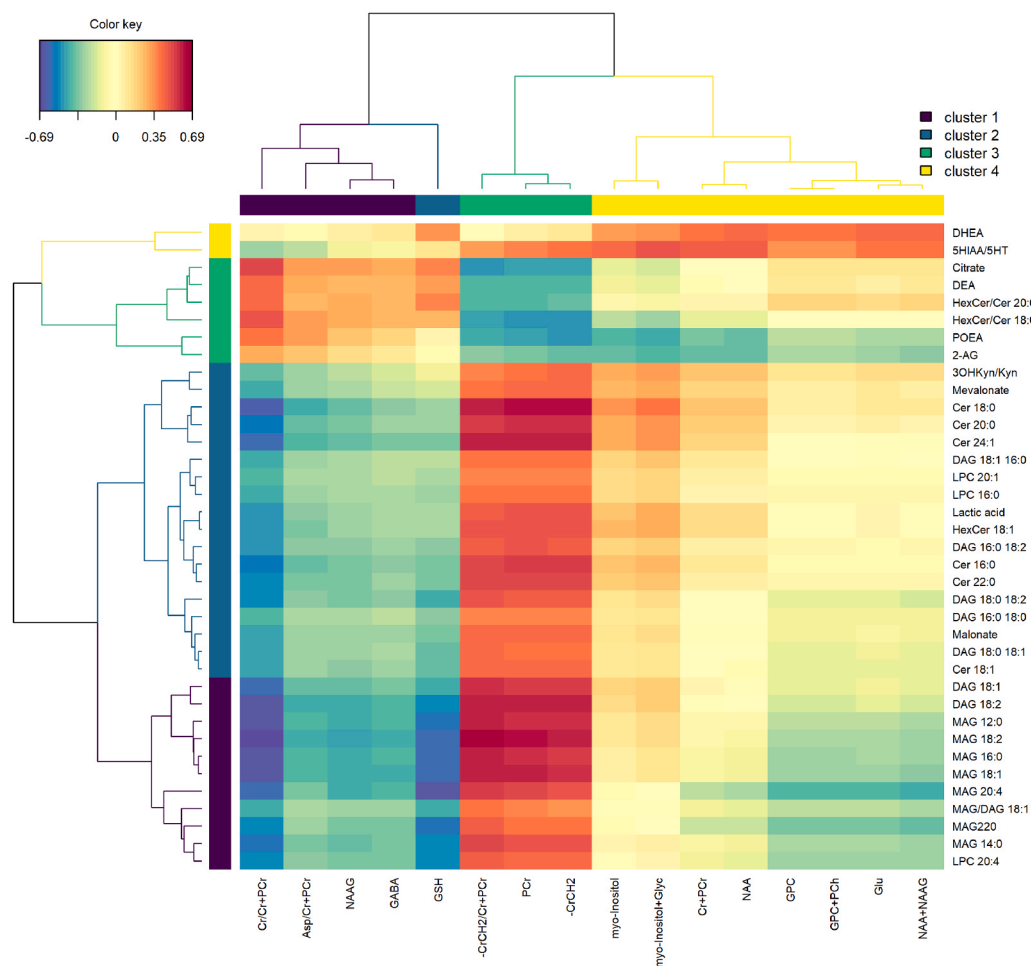


Fig. 4. Regularized canonical correlation analysis (rCCA) heatmap showing associations between plasma metabolite Δ -values and posterior cingulate cortex proton magnetic resonance spectroscopy (^1H -MRS) metabolite ratios. Rows represent plasma metabolites and columns represent posterior cingulate cortex MRS metabolites. Color intensity indicates the strength and direction of the regularized correlation coefficient. Hierarchical clustering was applied to rows and columns, and four column clusters are indicated above the heatmap. Abbreviations: 2-AG, 2-arachidonoylglycerol; 3OHKyn/Kyn, 3-hydroxykynurenine/kynurenine ratio; 5HIAA/5HT, 5-hydroxyindoleacetic acid/serotonin ratio; Asp, aspartate; Cer, ceramide; Cr, creatine; DAG, diacylglycerol; DEA, docosatraenoyl ethanolamide; DHEA, dehydroepiandrosterone; GABA, gamma-aminobutyric acid; Glu, glutamate; Glyc, glycine; GPC, glycerophosphocholine; GPC+PCh, glycerophosphocholine plus phosphocholine; GSH, glutathione; HexCer/Cer, hexosylceramide/ceramide ratio; LPC, lysophosphatidylcholine; NAA, N-acetylaspartate; NAA+NAAG, N-acetylaspartate plus N-acetylaspartylglutamate; NAAG, N-acetylaspartylglutamate; PCC, posterior cingulate cortex; PCr, phosphocreatine; POEA, palmitoleoyl ethanolamide.

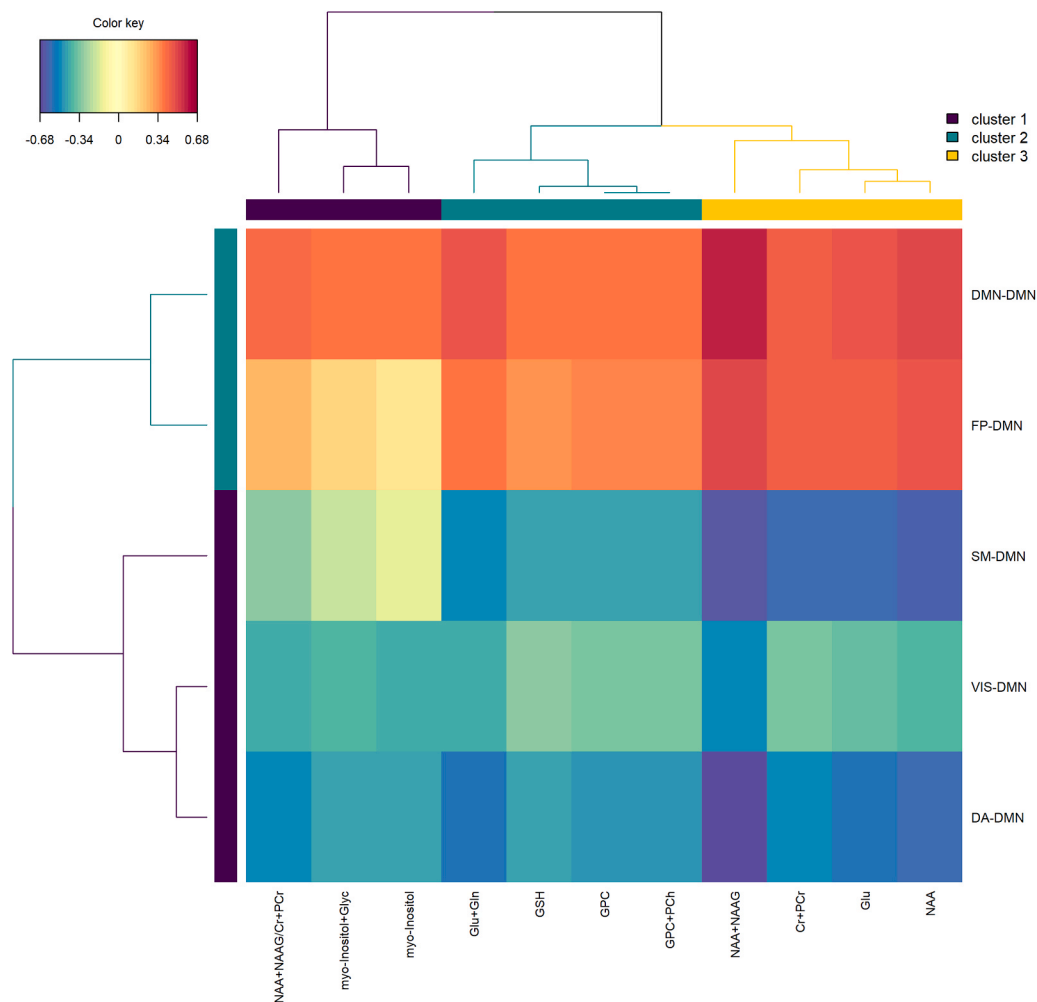


Fig. 5. Regularized canonical correlation analysis (rCCA) heatmap showing associations between posterior cingulate cortex proton magnetic resonance spectroscopy (^1H -MRS) metabolite ratios and default mode network (DMN)-related functional connectivity measures. Rows represent DMN-related functional connectivity pairs and columns represent posterior cingulate cortex MRS metabolites. Color intensity indicates the strength and direction of the regularized correlation coefficient. Hierarchical clustering was applied to rows and columns, and three column clusters are indicated above the heatmap. Abbreviations: Cr, creatine; DA, dorsal attention network; DMN, default mode network; FP, frontoparietal network; Gln, glutamine; Glu, glutamate; GPC, glycerophosphocholine; GPC+PCh, glycerophosphocholine plus phosphocholine; GSH, glutathione; Glyc, glycine; NAA, N-acetylaspartate; NAAG, N-acetylaspartylglutamate; PCr, phosphocreatine; SM, sensorimotor network; VIS, visual network.

related connectivity, while myo-inositol showed an inverse pattern relative to these metabolites. The corresponding regularized correlation matrix is provided in [Supplementary Table 17](#).

In the visual cortex, rCCA revealed generally weak associations between MRS-derived metabolites and connectivity measures. Only two modest relationships were observed, namely between GABA and visual-attention connectivity and between taurine and visual-frontoparietal connectivity (data not shown), without consistent patterns across variables. Several metabolites were excluded because of insufficient spectral quality, and one participant was removed. Given the limited and non-systematic nature of these findings, the visual cortex rCCA results were not interpreted further.

4. Discussion

This integrative analysis provides a multiscale view of the ayahuasca experience, linking peripheral biochemical changes, posterior cingulate cortex neurochemistry, functional brain network reconfiguration, and alterations in subjective consciousness. By modelling within-subject Δ -values across alkaloid concentrations, plasma metabolomics, resting-state functional connectivity, and 5D-ASC scores, we identified

biological shifts associated with key phenomenological dimensions of the psychedelic state. Specifically, OB, VUS, and AUD emerged as central experiential domains, each associated with distinct yet overlapping neurobiological signatures. The neurobiological fingerprints included all connectome networks, although edges involving the DMN and DA networks were overrepresented. In addition, they included metabolites related to energy metabolism, the endocannabinoid system, neurotransmission-related compounds such as glutamate and serotonin, and alkaloids present in the ayahuasca brew. Complementary rCCA analyses further supported a cautious interpretation of the peripheral findings by showing structured covariation between plasma metabolites, posterior cingulate cortex neurochemistry, and DMN-related functional connectivity. Several peripheral lipid species, including MAG-, DAG-, and ceramide-related measures, were inversely associated with posterior cingulate cortex markers such as creatine, aspartate, NAAG, and glutathione, whereas phosphocreatine showed the opposite pattern. In parallel, the plasma 5HIAA/serotonin ratio was positively associated with myo-inositol, NAA, and choline-containing compounds in the posterior cingulate cortex. A second rCCA indicated that posterior cingulate cortex metabolites including NAA, NAAG, and glutamate/glutamine were positively related to within-DMN and

frontoparietal–DMN connectivity, while showing the opposite pattern for dorsal attention–DMN, visual–DMN, and sensorimotor–DMN connectivity; myo-inositol showed an inverse profile. These findings do not imply that peripheral metabolites drive brain network changes, or vice versa, but they do suggest that ayahuasca-related peripheral metabolic responses are aligned with central neurochemical states that map onto the reconfiguration of core large-scale networks.

The prominence of associations between the DMN and DA networks and the psychedelic experience under ayahuasca is notable. The DMN supports internally directed thought, including self-reflection, autobiographical memory, and maintaining a coherent sense of self [40]. It is active when the mind is not focused on the external environment. In contrast, the DA network supports externally directed attention, helping to select and process sensory information relevant to current goals [41]. These two networks typically show anti-correlated activity, meaning that when one is active, the other is suppressed [41,42]. Their dynamic balance allows for flexible shifts between inner experience and engagement with the external world, forming a core organizational feature of normal waking consciousness. Under classic psychedelics, activity and functional connectivity within the DMN are reduced, particularly in hub regions such as the posterior cingulate cortex [14]. This is much in line with the observation in the current study that changes within the DMN were associated with changes in the psychedelic state. At the same time, the usual anti-correlation between the DMN and the DA weakened during the psychedelic state with ayahuasca, suggesting that internal and external modes of processing became less segregated. This seems in line with the general notion that under psychedelics, global network connectivity increases, producing a more entropic and integrated pattern of brain activity [13,43,44]. For example, although obtained with intravenous DMT rather than ayahuasca, recent multimodal EEG–fMRI work in humans showed convergent evidence of increased global functional connectivity, reduced network integrity, compression of the principal cortical gradient, and decreased alpha power, supporting the view that DMT-class psychedelic effects involve large-scale 5-HT_{2A}-linked reorganization of brain dynamics [45]. Subjectively, this may correspond to intensified sensory experience, novel associations, and altered self-perception that are characteristic of psychedelic states.

The integrative multiblock model revealed that these network reorganizations are biochemically coupled to specific lipid and metabolic pathways. The component explaining the largest shared variance linked OB, VUS, and AUD with circulating ayahuasca alkaloids, endocannabinoid-related N-acyl ethanolamines, energy metabolism intermediates, and lipid-derived metabolites, together with distributed changes in large-scale functional connectivity. Although these associations do not imply causality, the observed relationships with neurotransmission-related compounds such as serotonin and glutamate remain broadly consistent with canonical models of psychedelic action in which 5-HT_{2A} receptor activation engages glutamatergic cortical signalling [46–48].

Associations with energy metabolism and amino acid-related neurotransmitters may reflect coordinated physiological processes rather than direct mechanistic effects. While previous work suggests that psychedelics can influence mitochondrial and oxidative activity in ways that support synaptic reorganization [49–51], the relationships observed here may alternatively reflect adaptive metabolic responses accompanying neural plasticity, rather than drivers of it. In addition, these associations could arise from peripheral or systemic processes, potentially mediated by serotonergic mechanisms beyond 5-HT_{2A} [52], including interactions with receptors such as 5-HT_{2B}, which are expressed in peripheral tissues and linked to metabolic and cellular signalling pathways [53]. Thus, the observed metabolic signatures may reflect a combination of central neuroplastic processes and peripheral physiological responses, rather than a unidirectional effect of psychedelics on brain energy metabolism.

Ceramide and acylglycerol lipids however have received little

attention as relevant contributors to the psychedelic state before. Ceramides are bioactive lipids that modulate membrane microstructure, synaptic homeostasis, inflammatory tone, and are increasingly recognized as modulators of affective and cognitive states [54,55], while DAGs and MAGs represent key bioactive lipids in the endocannabinoid system [56,57]. Interestingly, the MAG lipids also included 2-AG (MAG 20:4) [56,58], reflecting the endocannabinoid lipid signalling during the psychedelic experience. DAGs are biochemical precursors involved in the synthesis of 2-AG. 2-AG signalling contributes to retrograde synaptic modulation, regulates excitatory–inhibitory neurotransmission, and shapes synaptic plasticity [59,60]. The dynamic interplay between DAG and MAG (including 2-AG) provides an activity-dependent feedback system that adjusts synaptic strength and excitability of neural circuits [58,61].

The negative relationships of psychedelic state features with DAG and MAG species suggest heightened activity of the 2-AG signalling cycle, where DAG is more rapidly converted to 2-AG by DAGL- α/β and MAG species reflect faster 2-AG breakdown [56]. These findings are also in line with reports of elevated 2-AG mobilization and neuroplasticity following serotonergic stimulation [62,63]. Conversely, positive relationships of HexCer/Cer 18:0 and 20:0 with psychedelic state and connectome features suggests that increments in ceramide metabolism are linked to synaptic network modulation and affective states [54]. These findings extend serotonergic and glutamatergic models by showing that lipid-signalling pathways, particularly those involving glycerolipids and sphingolipids, covary with network flexibility and may influence large-scale functional brain organization. The coordinated variation between these lipid species, circulating alkaloids, and functional connectivity changes aligns with evidence that endocannabinoid–serotonin crosstalk supports neurophysiological plasticity [64,65].

Complementary rCCA analyses further supported this interpretation by showing structured covariation between peripheral metabolites, posterior cingulate cortex neurochemistry, and DMN-related functional connectivity. In particular, several peripheral lipid species, including MAG-, DAG-, and ceramide-related measures, were inversely associated with posterior cingulate cortex neurochemical markers such as creatine, aspartate, NAAG, and glutathione, whereas phosphocreatine showed the opposite pattern. In parallel, the plasma 5HIAA/serotonin ratio was positively associated with myo-inositol, NAA, and choline-containing compounds in the posterior cingulate cortex. A second rCCA indicated that posterior cingulate cortex metabolites including NAA, NAAG, and glutamate/glutamine were positively related to within-DMN and frontoparietal–DMN connectivity, while showing the opposite pattern for dorsal attention–DMN, visual–DMN, and sensorimotor–DMN connectivity; myo-inositol showed an inverse profile. Although these associations are not causal, they suggest that peripheral lipid and monoamine-related shifts under ayahuasca are aligned with central neurochemical states that map onto reconfiguration of core large-scale networks.

When focusing on the overlapping network features linked to ASC dimensions, acylglycerols and ceramides emerged as the primary metabolic correlates, while neurotransmission-related compounds were no longer prominently represented. This pattern suggests a shift toward secondary biochemical processes downstream of receptor activation, with the shared neurobiological signature characterized more strongly by sphingolipid (ceramide) and glycerolipid (acylglycerol) signalling than by direct monoamine-related metabolites. The dominant connectome features remained stable and edges involving the DMN and DA network continued to represent central axes of network reorganization. The prominence of acylglycerols and ceramides in the shared ASC-related network suggests that the subjective psychedelic experience may not only depend on primary neurotransmitter release but also on downstream lipid-mediated processes that reorganize large-scale functional connectivity of brain networks. In this framework, receptor modulation by DMT and harmine may initiate biochemical cascades in lipid signalling pathways that stabilize or amplify neural reconfiguration, thereby potentially contributing to altered states of consciousness.

Overall, these findings support the notion that conscious experience is grounded in dynamic, system-wide integration rather than isolated neural events, incorporating peripheral processes in a manner consistent with contemporary neurophenomenological accounts of embodied consciousness [66,67].

While this study provides an unprecedented integrative view of ayahuasca's multiscale effects, several limitations should be acknowledged. First, the sample size was modest, reflecting the challenges of conducting multimodal imaging and biochemical analyses in naturalistic ceremonial contexts. Access to this type of biological material and highly synchronized multimodal data remains logistically and ethically complex, which constrains replication and large-scale validation efforts. Moreover, as all participants were experienced users, the present findings may reflect adaptations associated with repeated ayahuasca exposure. Responses in novice or clinical populations could differ in both direction and magnitude.

Second, while previous work suggests that psychedelics can influence mitochondrial and oxidative activity in ways that support synaptic reorganization [49–51], the relationships observed here may reflect adaptive metabolic responses accompanying neural plasticity, rather than its drivers. In addition, these associations could arise from peripheral or systemic processes, potentially mediated by serotonergic mechanisms beyond 5-HT_{2A} [52], including interactions with receptors such as 5-HT_{2B}, which are expressed in peripheral tissues and linked to metabolic and cellular signalling pathways [53]. Thus, the observed metabolic signatures may reflect a combination of central neuroplastic processes and peripheral physiological responses, rather than a unidirectional effect of psychedelics on brain energy metabolism. Although the additional rCCA analyses revealed structured covariation between peripheral metabolites, posterior cingulate cortex neurochemistry, and DMN-related functional connectivity, (Fig. 4; Supplementary Tables 16–17), these analyses remain correlational and do not establish causal directionality. Accordingly, the observed associations may reflect parallel responses to alkaloid exposure across peripheral and central systems, reciprocal interactions between them, or a combination of both.

Third, despite measuring plasma alkaloid levels at 60 min post drinking, the inherent complexity of the ayahuasca brew and the inter-individual variability of the amount of ayahuasca consumed may introduce variability in drug concentrations and interactions with peripheral/central metabolites and brain imaging measures that might be better captured in pharmacokinetic-pharmacodynamic models that include multiple measures over the entire time window of the ayahuasca experience. Methodologically, the cross-sectional approach captures acute effects but does not address temporal dynamics or longer-term adaptations. Future work should incorporate longitudinal sampling to track recovery, persistence, or plasticity in both peripheral metabolomic and neural connectome signatures. Finally, while the current integrative framework bridges peripheral and central processes, the experiential dimension remains inherently complex. Advancing this line of research will require combining computational modelling, real-time neural recordings, and fine-grained psychometric or phenomenological assessments to better map how molecular and network-level dynamics give rise to subjective experience. Such multimodal, systems-level approaches will be critical for elucidating the neurobiological foundations of consciousness and informing the therapeutic potential of psychedelics.

In conclusion, the present findings indicate that the acute effects of ayahuasca reflect a multiscale brain-body reorganization linking alkaloid exposure, peripheral metabolism, posterior cingulate cortex neurochemistry, large-scale functional connectivity, and conscious experience. Beyond established serotonergic models, our results identify endocannabinoid- and lipid-related pathways, particularly acylglycerols and ceramides, as prominent correlates of psychedelic phenomenology and network reconfiguration. The observed alignment between peripheral metabolites, posterior cingulate cortex neurochemical profiles, and

DMN-related connectivity further supports the value of integrative multimodal approaches for characterizing how altered states of consciousness emerge from coordinated interactions across molecular, regional, and systems-level domains. This work therefore advances ayahuasca research from a primarily receptor-centred account toward a broader systems neurobiology of psychedelic experience.

CRedit authorship contribution statement

Francisco Madrid-Gambin: Writing – original draft, Visualization, Validation, Formal analysis, Conceptualization. **Pablo Mallaroni:** Methodology, Formal analysis, Data curation. **Johannes G. Ramaekers:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Kim van Oorsouw:** Formal analysis. **Stefan W. Toennes:** Formal analysis. **Lilian Kloft-Heller:** Methodology. **Natasha L. Mason:** Methodology, Formal analysis. **Johannes T. Reckweg:** Formal analysis. **Noemí Haro:** Formal analysis. **Oscar J. Pozo:** Formal analysis, Conceptualization.

Funding

JR acknowledges financial support from Dutch Research Council (NWO, grant number 406.18. GO.019).

Declaration of Competing Interest

None.

Acknowledgments

We are grateful for the extended cooperation of members of the church of Santo Daime.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2026.119391.

Data availability

Data will be made available on request.

References

- [1] G.H. de M. Santos, L.S. Rodrigues, J.M. Rocha, G.N. Rossi, G. Ona, J.C. Bouso, J.E. C. Hallak, R.G. dos Santos, Neural network modulation of ayahuasca: a systematic review of human studies, *Psychoactives* 2 (2023) 76–91, <https://doi.org/10.3390/PSYCHOACTIVES2010006>.
- [2] J. Riba, M. Valle, G. Urbano, M. Yritia, A. Morte, M.J. Barbanoj, Human pharmacology of ayahuasca: subjective and cardiovascular effects, monoamine metabolite excretion, and pharmacokinetics, *J. Pharmacol. Exp. Ther.* 306 (2003) 73–83, <https://doi.org/10.1124/JPET.103.049882>.
- [3] A. Ábelo, J.W. Smallridge, R. von Rotz, D.A. Dornbierer, K. Egger, M. Ashton, M. Scheidegger, Population pharmacokinetic-pharmacodynamic modeling of co-administered N,N-dimethyltryptamine and harmine in healthy subjects, *Biomed. Pharm.* 189 (2025) 118329, <https://doi.org/10.1016/j.biopha.2025.118329>.
- [4] F. Madrid-Gambin, A. Gomez-Gomez, A. Busquets-García, N. Haro, S. Marco, N. L. Mason, J.T. Reckweg, P. Mallaroni, L. Kloft, K. van Oorsouw, S.W. Toennes, R. de la Torre, J.G. Ramaekers, O.J. Pozo, Metabolomics and integrated network analysis reveal roles of endocannabinoids and large neutral amino acid balance in the ayahuasca experience, *Biomed. Pharm.* 149 (2022) 112845, <https://doi.org/10.1016/J.BIOPHA.2022.112845>.
- [5] R.L. Smith, H. Canton, R.J. Barrett, Agonist properties of N,N-dimethyltryptamine at serotonin 5-HT_{2A} and 5-HT_{2C} receptors, *Pharmacol. Biochem. Behav.* 61 (1998) 323–330, [https://doi.org/10.1016/S0091-3057\(98\)00110-5](https://doi.org/10.1016/S0091-3057(98)00110-5).
- [6] A.A. Nunes, R.G. dos Santos, F.L. Osório, R.F. Sanches, J.A.S. Crippa, J.E.C. Hallak, Effects of Ayahuasca and its Alkaloids on Drug Dependence: A Systematic Literature Review of Quantitative Studies in Animals and Humans, (<http://Dx.Doi.Org/10.1080/02791072.2016.1188225>) 48 (2016) 195–205. <https://doi.org/10.1080/02791072.2016.1188225>.
- [7] G.N. Rossi, L.T.L. Guerra, G.B. Baker, S.M. Dursun, J.C.B. Saiz, J.E.C. Hallak, R. G. dos Santos, Molecular pathways of the therapeutic effects of ayahuasca, a

- botanical psychedelic and potential rapid-acting antidepressant, *Biomolecules* 12 (2022), <https://doi.org/10.3390/biom12111618>.
- [8] R.G. Dos Santos, J.E.C. Hallak, Ayahuasca: pharmacology, safety, and therapeutic effects, *CNS Spectr.* 30 (2025) e2, <https://doi.org/10.1017/S109285292400213X>.
- [9] M. Valle, A.E. Maqueda, M. Rabella, A. Rodríguez-Pujadas, R.M. Antonijoan, S. Romero, J.F. Alonso, M.A. Mañanas, S. Barker, P. Friedlander, A. Feilding, J. Riba, Inhibition of alpha oscillations through serotonin-2A receptor activation underlies the visual effects of ayahuasca in humans, *Eur. Neuropsychopharmacol.* 26 (2016) 1161–1175, <https://doi.org/10.1016/j.euroneuro.2016.03.012>.
- [10] N. Hesselgrave, T.A. Troppoli, A.B. Wulff, A.B. Cole, S.M. Thompson, Harnessing psilocybin: antidepressant-like behavioral and synaptic actions of psilocybin are independent of 5-HT_{2R} activation in mice, *Proc. Natl. Acad. Sci.* 118 (2021), <https://doi.org/10.1073/pnas.2022489118>.
- [11] F. Madrid-Gambin, D. Fabregat-Safont, A. Gomez-Gomez, E. Olesti, N.L. Mason, J. G. Ramaekers, O.J. Pozo, Present and future of metabolic and metabolomics studies focused on classical psychedelics in humans, *Biomed. Pharm.* 169 (2023) 115775, <https://doi.org/10.1016/j.biopha.2023.115775>.
- [12] F. Madrid-Gambin, N. Haro, N.L. Mason, P. Mallaroni, E.L. Theunissen, S. W. Toennes, O.J. Pozo, J.G. Ramaekers, Metabolomic profiling of cannabis use and cannabis intoxication in humans, 2025 506, *Neuropsychopharmacol* 50 (2025) 920–927, <https://doi.org/10.1038/s41386-025-02082-7>.
- [13] J.S. Siegel, S. Subramanian, D. Perry, B.P. Kay, E.M. Gordon, T.O. Laumann, T. R. Reneau, N.V. Metcalf, R.V. Chacko, C. Gratton, C. Horan, S.R. Kimmel, J. S. Shimony, J.A. Schweiger, D.F. Wong, D.A. Bender, K.M. Scheidter, F.I. Whiting, J.A. Padawer-Curry, R.T. Shinohara, Y. Chen, J. Moser, E. Yacoub, S.M. Nelson, L. Vizioli, D.A. Fair, E.J. Lenze, R. Carhart-Harris, C.L. Raison, M.E. Raichle, A. Z. Snyder, G.E. Nicol, N.U.F. Dosenbach, Psilocybin desynchronizes the human brain, 2024 6328023, *Nat* 632 (2024) 131–138, <https://doi.org/10.1038/s41586-024-07624-5>.
- [14] J.J. Gattuso, D. Perkins, S. Ruffell, A.J. Lawrence, D. Hoyer, L.H. Jacobson, C. Timmermann, D. Castle, S.L. Rossell, L.P. Downey, B.A. Pagni, N.L. Galvão-Coelho, D. Nutt, J. Sarris, Default mode network modulation by psychedelics: a systematic review, *Int. J. Neuropsychopharmacol.* 26 (2023) 155–188, <https://doi.org/10.1093/ijnp/pyac074>.
- [15] A.V. Lebedev, M. Kaelin, M. Lövdén, J. Nilsson, A. Feilding, D.J. Nutt, R.L. Carhart-Harris, LSD-induced entropic brain activity predicts subsequent personality change, *Hum. Brain Mapp.* 37 (2016) 3203–3213, <https://doi.org/10.1002/HBM.23234>.
- [16] M. Avram, L. Fortea, L. Wollner, R. Coenen, A. Korda, H. Rogg, F. Holze, P. Vizeili, L. Ley, J. Radua, F. Müller, M.E. Liechti, S. Borgwardt, Large-scale brain connectivity changes following the administration of lysergic acid diethylamide, d-amphetamine, and 3,4-methylenedioxymphetamine, *Mol. Psychiatry* 30 (2025) 1297–1307, <https://doi.org/10.1038/s41380-024-02734-Y;TECHMETA>.
- [17] P. Mallaroni, N.L. Mason, L. Klotz, J.T. Reckweg, K. van Oorsouw, S.W. Toennes, H. M. Tolle, E. Amico, J.G. Ramaekers, Shared functional connectome fingerprints following ritualistic ayahuasca intake, *Neuroimage* 285 (2024) 124080, <https://doi.org/10.1016/j.neuroimage.2023.120480>.
- [18] J.G. Ramaekers, P. Mallaroni, L. Klotz, J.T. Reckweg, S.W. Toennes, K. van Oorsouw, N.L. Mason, Altered state of consciousness and mental imagery as a function of N, N-dimethyltryptamine concentration in ritualistic Ayahuasca users, *J. Cogn. Neurosci.* 35 (2023) 1382–1393, https://doi.org/10.1162/jocn_a.02003.
- [19] A. Dittrich, The standardized psychometric assessment of altered states of consciousness (ASCs) in humans, *Pharmacopsychiatry* 31 (1998) 80–84, <https://doi.org/10.1055/s-2007-979351>.
- [20] A. Dittrich, D. Lamparter, M. Maurer, 5D-ASC: Questionnaire for the Assessment of Altered States of Consciousness. A Short Introduction 3rd Edn, Zürich PSIN PLUS, 2010.
- [21] E. Studerus, A. Gamma, F.X. Vollenweider, Psychometric evaluation of the altered states of consciousness rating scale (OAV), *PLoS One* 5 (2010) e12412, <https://doi.org/10.1371/journal.pone.0012412>.
- [22] H.M. Tolle, J.C. Farah, P. Mallaroni, N.L. Mason, J.G. Ramaekers, E. Amico, The unique neural signature of your trip: functional connectome fingerprints of subjective psilocybin experience, *Netw. Neurosci.* 8 (2024) 203, https://doi.org/10.1162/NETN_A_00349.
- [23] E. Amico, D. Marinazzo, C. Di Perri, L. Heine, J. Annen, C. Martial, M. Dzemidzic, M. Kirsch, V. Bonhomme, S. Laureys, J. Goñi, Mapping the functional connectome traits of levels of consciousness, *Neuroimage* 148 (2017) 201–211, <https://doi.org/10.1016/j.neuroimage.2017.01.020>.
- [24] E. Amico, M. Dzemidzic, B.G. Oberlin, C.R. Carron, J. Harezlak, J. Goñi, D. A. Kareken, The disengaging brain: Dynamic transitions from cognitive engagement and alcoholism risk, *Neuroimage* 209 (2020) 116515, <https://doi.org/10.1016/j.neuroimage.2020.116515>.
- [25] Y. Behzadi, K. Restom, J. Liao, T.T. Liu, A component based noise correction method (CompCor) for BOLD and perfusion based fMRI, *Neuroimage* 37 (2007) 90–101, <https://doi.org/10.1016/j.neuroimage.2007.04.042>.
- [26] J.D. Power, A. Mitra, T.O. Laumann, A.Z. Snyder, B.L. Schlaggar, S.E. Petersen, Methods to detect, characterize, and remove motion artifact in resting state fMRI, *Neuroimage* 84 (2014) 320–341, <https://doi.org/10.1016/j.neuroimage.2013.08.048>.
- [27] A. Schaefer, R. Kong, E.M. Gordon, T.O. Laumann, X.-N. Zuo, A.J. Holmes, S. B. Eickhoff, B.T.T. Yeo, Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI, *Cereb. Cortex* 28 (2018) 3095–3114, <https://doi.org/10.1093/CERCOR/BHX179>.
- [28] B.T. Thomas Yeo, F.M. Krienen, J. Sepulcre, M.R. Sabuncu, D. Lashkari, M. Hollinshead, J.L. Roffman, J.W. Smoller, L. Zöllei, J.R. Polimeni, B. Fisch, H. Liu, R.L. Buckner, The organization of the human cerebral cortex estimated by intrinsic functional connectivity, *J. Neurophysiol.* 106 (2011) 1125–1165, <https://doi.org/10.1152/JN.00338.2011>.
- [29] J. Frahm, H. Bruhn, M.L. Gyngell, K.D. Merboldt, W. Hänicke, R. Sauter, Localized high-resolution proton NMR spectroscopy using stimulated echoes: initial applications to human brain in vivo, *Magn. Reson. Med.* 9 (1989) 79–93, <https://doi.org/10.1002/MRM.1910090110>.
- [30] J. Near, A.D. Harris, C. Juchem, R. Kreis, M. Marjańska, G. Öz, J. Slotboom, M. Wilson, C. Gasparovic, Preprocessing, analysis and quantification in single-voxel magnetic resonance spectroscopy: experts' consensus recommendations, *NMR Biomed.* 34 (2021), <https://doi.org/10.1002/NBM.4257>.
- [31] A. Gomez-Gomez, E. Olesti, B. Montero-San-Martin, A. Soldevila, T. Deschamps, N. Pizarro, R. de la Torre, O.J. Pozo, Determination of up to twenty carboxylic acid containing compounds in clinically relevant matrices by o-benzylhydroxylamine derivatization and liquid chromatography-tandem mass spectrometry, *J. Pharm. Biomed. Anal.* 208 (2022) 114450, <https://doi.org/10.1016/j.jpba.2021.114450>.
- [32] E. Olesti, J. Rodríguez-Morató, A. Gomez-Gomez, J.G. Ramaekers, R. De La Torre, O.J. Pozo, Quantification of endogenous neurotransmitters and related compounds by liquid chromatography coupled to tandem mass spectrometry, (2018). <https://doi.org/10.1016/j.talanta.2018.09.034>.
- [33] J. Marcos, N. Renau, O. Valverde, G. Aznar-Lain, I. Gracia-Rubio, M. Gonzalez-Sepulveda, L.A. Pérez-Jurado, R. Ventura, J. Segura, O.J. Pozo, Targeting tryptophan and tyrosine metabolism by liquid chromatography tandem mass spectrometry, *J. Chromatogr. A* 1434 (2016) 91–101, <https://doi.org/10.1016/j.chroma.2016.01.023>.
- [34] A. Pastor, M. Farré, M. Fitó, F. Fernandez-Aranda, R. De La Torre, Analysis of ECs and related compounds in plasma: artifactual isomerization and ex vivo enzymatic generation of 2-MGs, *J. Lipid Res.* 55 (2014) 966–977, <https://doi.org/10.1194/JLR.D043794>.
- [35] A. Gomez-Gomez, J. Miranda, G. Feixas, A. Arranz Betegon, F. Crispi, E. Gratacós, O.J. Pozo, Determination of the steroid profile in alternative matrices by liquid chromatography tandem mass spectrometry, *J. Steroid Biochem. Mol. Biol.* 197 (2020), <https://doi.org/10.1016/j.jsbmb.2019.105520>.
- [36] J. Marín-Corral, J. Rodríguez-Morató, A. Gomez-Gomez, S. Pascual-Guardia, R. Muñoz-Bermúdez, A. Salazar-Degracia, P. Pérez-Terán, M.I. Restrepo, O. Khymentets, N. Haro, J.R. Masclans, O.J. Pozo, Metabolic signatures associated with severity in hospitalized COVID-19 patients, *Int. J. Mol. Sci.* 22 (2021), <https://doi.org/10.3390/IJMS22094794>.
- [37] A. Singh, C.P. Shannon, B. Gautier, F. Rohart, M. Vacher, S.J. Tebbutt, K.A. L. Cao, DIABLO: an integrative approach for identifying key molecular drivers from multi-omics assays, *Bioinformatics* 35 (2019) 3055–3062, <https://doi.org/10.1093/bioinformatics/bty1054>.
- [38] F. Rohart, B. Gautier, A. Singh, K.-A. Lê Cao, mixOmics: an R package for 'omics feature selection and multiple data integration, *PLoS Comput. Biol.* 13 (2017) e1005752, <https://doi.org/10.1371/journal.pcbi.1005752>.
- [39] C. Froidevaux, M.-L. Martin-Magniette, G. Rigai, Biological Data Integration: Computer and Statistical Approaches, (2024). (https://books.google.com/books/about/Biological_Data_Integration.html?hl=es&id=W9TnEAAQBAJ) (Accessed September 29, 2025).
- [40] V. Menon, 20 years of the default mode network: a review and synthesis, *Neuron* 111 (2023) 2469–2487, <https://doi.org/10.1016/j.neuron.2023.04.023>.
- [41] M.L. Dixon, J.R. Andrews-Hanna, R.N. Spreng, Z.C. Irving, C. Mills, M. Girn, K. Christoff, Interactions between the default network and dorsal attention network vary across default subsystems, time, and cognitive states, *Neuroimage* 147 (2017) 632–649, <https://doi.org/10.1016/j.neuroimage.2016.12.073>.
- [42] M.M. Owens, D.K. Yuan, S. Hahn, M. Albaugh, N. Allgaier, B. Chaarani, A. Potter, H. Garavan, Investigation of psychiatric and neuropsychological correlates of default mode network and dorsal attention network anticorrelation in children, *Cereb. Cortex* 30 (2020) 6083–6096, <https://doi.org/10.1093/CERCOR/BHAA143>.
- [43] R.L. Carhart-Harris, K.J. Friston, REBUS and the anarchic brain: toward a unified model of the brain action of psychedelics, *Pharmacol. Rev.* 71 (2019) 316–344, <https://doi.org/10.1124/PR.118.017160/-/DC1>.
- [44] J. Vohryzek, J. Cabral, C. Timmermann, S. Atasoy, L. Roseman, D.J. Nutt, R. L. Carhart-Harris, G. Deco, M.L. Kringsbach, The flattening of spacetime hierarchy of the N,N-dimethyltryptamine brain state is characterized by harmonic decomposition of spacetime (HADES) framework, *Natl. Sci. Rev.* 11 (2024), <https://doi.org/10.1093/NSR/NWAE124>.
- [45] C. Timmermann, L. Roseman, S. Haridas, F.E. Rosas, L. Luan, H. Kettner, J. Martell, D. Erritzoe, E. Tagliazucchi, C. Pallavicini, M. Girn, A. Alamaia, R. Leech, D.J. Nutt, R.L. Carhart-Harris, Human brain effects of DMT assessed via EEG-fMRI, *Proc. Natl. Acad. Sci.* 120 (2023), <https://doi.org/10.1073/PNAS.2218949120>.
- [46] F.X. Vollenweider, K.H. Preller, Psychedelic drugs: neurobiology and potential for treatment of psychiatric disorders, 2020 2111, *Nat. Rev. Neurosci.* 21 (2020) 611–624, <https://doi.org/10.1038/s41583-020-0367-2>.
- [47] N.L. Mason, K.P.C. Kuypers, F. Müller, J. Reckweg, D.H.Y. Tse, S.W. Toennes, N.R. P.W. Hutten, J.F.A. Jansen, P. Stiers, A. Feilding, J.G. Ramaekers, Me, myself, bye: regional alterations in glutamate and the experience of ego dissolution with psilocybin, *Neuropsychopharmacology* 45 (2020) 2003–2011, <https://doi.org/10.1038/s41386-020-0718-8>.
- [48] D.E. Nichols, Psychedelics, *Pharmacol. Rev.* 68 (2016) 264–355, <https://doi.org/10.1124/PR.115.011478>.
- [49] D.A. Martin, C.D. Nichols, The effects of hallucinogens on gene expression, in: A. L. Halberstadt, F.X. Vollenweider, D.E. Nichols (Eds.), *Behav. Neurobiol. Psychedelic Drugs*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2018, pp. 137–158, https://doi.org/10.1007/7854_2017_479.

- [50] M.P. Papageorgiou, M.D. Filiou, Mitochondrial dynamics and psychiatric disorders: the missing link, *Neurosci. Biobehav. Rev.* 165 (2024) 105837, <https://doi.org/10.1016/J.NEUBIOREV.2024.105837>.
- [51] P. Fissler, A. Vandersmissen, M. Filippi, R.N. Mavioglu, F. Scholkmann, A. Karabatsiakos, R. Krähenmann, Effects of serotonergic psychedelics on mitochondria: transdiagnostic implications for mitochondria-related pathologies, *J. Psychopharmacol.* 37 (2023) 679–686, <https://doi.org/10.1177/02698811231164707>;SUBPAGE:STRING:ABSTRACT;REQUESTEDJOURNAL: JOURNAL:JOPA;WEBSITE:WEBSITE:SAGE;ISSUE:ISSUE:DOI.
- [52] M.J. Keiser, V. Setola, J.J. Irwin, C. Lagner, A.I. Abbas, S.J. Hufeisen, N.H. Jensen, M.B. Kuijter, R.C. Matos, T.B. Tran, R. Whaley, R.A. Glennon, J. Hert, K.L. H. Thomas, D.D. Edwards, B.K. Shoichet, B.L. Roth, Predicting new molecular targets for known drugs, 2009 4627270, *Nat* 462 (2009) 175–181, <https://doi.org/10.1038/nature08506>.
- [53] K. Waters, Pharmacologic similarities and differences among hallucinogens, *J. Clin. Pharm.* 61 (2021) S100–S113, <https://doi.org/10.1002/jcph.1917>.
- [54] Y.A. Hannun, L.M. Obeid, Principles of bioactive lipid signalling: lessons from sphingolipids, *Nat. Rev. Mol. Cell Biol.* 9 (2008) 139–150, <https://doi.org/10.1038/NRM2329>;KWRD.
- [55] A. Dinoff, N. Herrmann, K.L. Lancôt, Ceramides and depression: a systematic review, *J. Affect. Disord.* 213 (2017) 35–43, <https://doi.org/10.1016/J.JAD.2017.02.008>.
- [56] D. Piomelli, The molecular logic of endocannabinoid signalling, *Nat. Rev. Neurosci.* 4 (2003) 873–884, <https://doi.org/10.1038/NRN1247>;KWRD.
- [57] T. Sugiura, S. Kishimoto, S. Oka, M. Gokoh, Biochemistry, pharmacology and physiology of 2-arachidonoylglycerol, an endogenous cannabinoid receptor ligand, *Prog. Lipid Res.* 45 (2006) 405–446, <https://doi.org/10.1016/j.plipres.2006.03.003>.
- [58] J.L. Blankman, G.M. Simon, B.F. Cravatt, A comprehensive profile of brain enzymes that hydrolyze the endocannabinoid 2-arachidonoylglycerol, *Chem. Biol.* 14 (2007) 1347–1356, <https://doi.org/10.1016/J.CHEMBIOL.2007.11.006>.
- [59] P.E. Castillo, T.J. Younts, A.E. Chávez, Y. Hashimoto, Endocannabinoid signaling and synaptic function, *Neuron* 76 (2012) 70–81, <https://doi.org/10.1016/J.NEURON.2012.09.020>.
- [60] I. Katona, T.F. Freund, Multiple functions of endocannabinoid signaling in the brain, *Annu. Rev. Neurosci.* 35 (2012) 529–558, <https://doi.org/10.1146/ANNUREV-NEURO-062111-150420>.
- [61] M. Kano, T. Ohno-Shosaku, Y. Hashimoto, M. Uchigashima, M. Watanabe, Endocannabinoid-Mediated Control of Synaptic Transmission, (<https://doi.org/10.1152/Physrev.00019.2008>) 89 (2009) 309–380. <https://doi.org/10.1152/PHYSREV.00019.2008>.
- [62] B. Basavarajappa, Neuropharmacology of the endocannabinoid signaling system-molecular mechanisms, biological actions and synaptic plasticity, *Curr. Neuropharmacol.* 5 (2007) 81–97, <https://doi.org/10.2174/157015907780866910>.
- [63] B. Bystrowska, I. Smaga, M. Frankowska, M. Filip, Changes in endocannabinoid and N-acyl ethanolamine levels in rat brain structures following cocaine self-administration and extinction training, *Prog. Neuro-Psychopharmacol. Biol. Psychiatry* 50 (2014) 1–10, <https://doi.org/10.1016/j.pnpbp.2013.12.002>.
- [64] C.J. Hillard, The endocannabinoid signaling system in the CNS: a primer, *Int. Rev. Neurobiol.* 125 (2015) 1–47, <https://doi.org/10.1016/BS.IRN.2015.10.001>.
- [65] M. Morena, S. Patel, J.S. Bains, M.N. Hill, Neurobiological interactions between stress and the endocannabinoid system, *Neuropsychopharmacology* 41 (2016) 80–102, <https://doi.org/10.1038/NPP.2015.166>.
- [66] F.J. Varela, Neurophenomenology: a methodological remedy for the hard problem, *J. Conscious. Stud.* 3 (1996) 330–349.
- [67] R. Millière, R.L. Carhart-Harris, L. Roseman, F.M. Trautwein, A. Berkovich-Ohana, Psychedelics, meditation, and self-consciousness, *Front. Psychol.* 9 (2018), <https://doi.org/10.3389/FPSYG.2018.01475>.